

Josh Girardini

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Summary

Senior Data Engineer building real-time data platforms for fintech and e-commerce systems. Hands-on experience with **Apache Kafka**, **Spark Structured Streaming**, **Scala**, and **Python** developing high-volume event pipelines and financial transaction processing systems. Built distributed data pipelines on **AWS** using **Amazon MSK**, **Amazon EMR**, **Amazon S3**, and **Apache Airflow**. Experience implementing **CDC pipelines** with **Kafka Connect** and **Debezium**, and developing **fraud monitoring** and **risk analytics** datasets on large-scale **Spark** data platforms.

Work Experience

Dave, Torrance, CA
Lead Data Engineer

Sep 2024 – Present

- Worked on building a real-time financial event streaming platform on **AWS** using **Apache Kafka (Amazon MSK)**, **Spark Structured Streaming**, **Amazon EMR**, and **Amazon S3**, ingesting payment transactions, mobile telemetry, and backend microservice events from the Dave banking platform.
- Built streaming pipelines in **Scala Spark** using **Spark SQL**, **Spark DataFrame APIs**, and **Spark Structured Streaming** to process high-volume **Kafka event streams** and generate transaction aggregates and sessionized activity datasets stored as **Parquet** datasets in **Amazon S3**.
- Set up **Kafka topics**, **partition strategies**, **consumer groups**, and replication settings on **Amazon MSK** to support reliable ingestion of financial transaction events and application activity streams.
- Implemented **CDC pipelines** from **PostgreSQL** using **Kafka Connect** and **Debezium**, streaming **WAL change events** into **Kafka topics** so downstream pipelines could process transaction updates in near real time.
- Converted several **batch ETL workflows** into **streaming pipelines** using **Spark Structured Streaming**, improving data availability for transaction monitoring and internal financial analytics systems.
- Worked on a **fraud monitoring streaming pipeline** using **Spark Structured Streaming**, **PySpark**, and **Spark window functions**, generating fraud signals such as **transaction velocity**, abnormal spending behavior, and unusual merchant activity.
- Built **feature engineering pipelines** in **PySpark** and **Spark SQL** generating behavioral datasets used by internal **risk scoring** and **fraud monitoring** systems.
- Maintained **Apache Airflow DAGs** orchestrating **Spark jobs**, **CDC ingestion tasks**, and data backfills, while monitoring pipeline health using **Prometheus**, **Grafana**, and **AWS CloudWatch** for **Kafka brokers**, **Spark executors**, and **EMR clusters**.

Dave, Torrance, CA
Senior Data Engineer

Jan 2022 – Sep 2024

- Worked on the financial transaction data platform on **AWS** using **Apache Kafka (Amazon MSK)**, **Amazon S3**, **Amazon EMR**, and **PostgreSQL**, ingesting banking activity from **Plaid APIs**, payment processors, and internal financial services.

- Built ingestion pipelines in **Python** handling **REST API integrations**, **JSON transaction payloads**, and publishing normalized financial events to **Kafka topics** using **Avro schemas** and **Confluent Schema Registry**.
- Developed distributed **Spark ETL pipelines** using **Spark SQL**, **PySpark**, and **Spark DataFrame APIs** on **EMR clusters**, transforming raw transaction logs into structured financial datasets stored as partitioned **Parquet tables** in **Amazon S3**.
- Worked on a **fraud monitoring data pipeline** processing **Kafka transaction streams**, generating fraud signals using **Spark aggregations**, **window functions**, and transaction frequency metrics.
- Built and maintained **Apache Airflow DAGs** scheduling ingestion workflows, **Spark batch jobs**, and **data reconciliation pipelines** across transaction and account datasets.
- Improved pipeline performance through **PostgreSQL table partitioning**, **B-tree indexing**, **Spark executor tuning**, **S3 partition pruning**, and **EMR autoscaling**, supporting large-scale financial transaction processing.
- Maintained data validation and monitoring across pipelines using **Airflow task logging**, **CloudWatch metrics**, and pipeline-level data quality checks.

Thrive Market, Marina del Rey, CA
Senior Data Engineer

May 2020 – Jan 2022

- Built a customer behavior event data platform on **AWS** using **Amazon Kinesis Data Streams**, **Amazon S3**, and **EC2 ingestion services**, collecting web and mobile events including product views, search activity, cart updates, and checkout flows.
- Developed distributed data processing pipelines using **Apache Spark (PySpark, Spark SQL)** on **Amazon EMR**, transforming raw JSON event streams into structured datasets stored as **partitioned Parquet tables in Amazon S3 data lake storage**.
- Implemented event ingestion pipelines integrating **Segment tracking events**, **REST APIs**, and **mobile telemetry**, publishing event payloads into **Kinesis streams** and **S3 raw data layers**, and generating datasets using **Spark aggregations**, **window functions**, and **sessionization logic** for customer engagement and subscription lifecycle analytics.
- Maintained **Apache Airflow DAGs running on EC2** orchestrating **Spark batch jobs**, **ingestion services**, and **S3 data lake processing workflows**, monitoring pipeline health using **AWS CloudWatch logs and metrics**.
- Delivered curated analytics datasets loaded into **Amazon Redshift**, supporting **product analytics**, **conversion funnel monitoring**, and **retention analysis** used by internal analytics teams and **Tableau dashboards**.

Thrive Market, Marina del Rey, CA
Senior Analytics Engineer

Jun 2019 – May 2020

- Built the **Marketing Attribution & Customer Acquisition Analytics Platform** on **AWS** using **Amazon S3**, **Amazon Redshift**, **AWS Glue**, and **Python**, integrating marketing campaign data, customer activity events, and e-commerce order transactions.
- Developed ingestion pipelines using **Python ETL scripts**, **REST APIs**, and **scheduled batch jobs**, collecting datasets from **Google Ads**, **Facebook Ads**, **Klaviyo campaigns**, and **internal order systems**, staging raw datasets in **Amazon S3 data lake storage**.
- Designed analytics data models in **Amazon Redshift** using complex **SQL joins**, **CTEs**, **window functions**, and **aggregations**, generating datasets for **multi-touch attribution**, **conversion funnels**, **CAC**, and **ROAS metrics**.
- Built transformation workflows loading datasets from **S3 to Redshift** using **COPY commands**, optimizing performance with **sort keys**, **distribution keys**, **compression encoding**, and **query tuning**.

- Maintained **Apache Airflow DAGs on EC2** orchestrating **Python ETL pipelines, AWS Glue jobs, S3 data processing, and Redshift refresh workflows**, delivering datasets used by **Tableau dashboards** for campaign performance and acquisition analytics.

Thrive Market, Marina del Rey, CA
Analytics Engineer

Aug 2018 – Jun 2019

- Built analytics datasets on **AWS** using **Amazon S3, Amazon Redshift, and Python**, processing e-commerce order transactions, product catalog data, and customer activity logs from the Thrive Market platform.
- Developed **SQL data models in Amazon Redshift** using joins, aggregations, CTEs, and window functions to generate reporting tables for **marketing attribution, order analytics, and product performance metrics**.
- Maintained **data ingestion and transformation workflows** using **Python ETL scripts, Redshift COPY commands, and S3 batch loads**, preparing datasets used by analysts and **Tableau dashboards** for marketing and operations reporting.

RAND Corporation, Santa Monica, CA
Research Assistant

Jul 2015 – May 2018

- Worked with healthcare research datasets on **Google Cloud Platform (GCP)** using **BigQuery, Google Cloud Storage (GCS), and Cloud Dataproc**, preparing patient records, claims data, and treatment datasets for research analysis.
- Used **Python, SQL, and PySpark on Dataproc clusters** to clean and normalize healthcare datasets, writing processed outputs to **BigQuery tables and Parquet datasets stored in GCS**.
- Wrote **BigQuery SQL queries** using joins, aggregations, and window functions to generate cohort datasets and healthcare utilization metrics used by RAND healthcare policy research teams.

Education

Bachelor of Arts, Harvard University, Cambridge, MA

2011 - 2015

Top Skills

- **Programming Languages:** Python, Scala, SQL, R, Bash
- **Streaming & Distributed Processing:** Apache Kafka, Amazon MSK, Kafka Connect, Debezium, Confluent Schema Registry, Apache Spark, Spark Structured Streaming, PySpark, Spark SQL
- **Data Engineering & Pipelines:** Event-Driven Data Pipelines, Streaming Data Pipelines, CDC Pipelines, ETL / ELT Pipelines, Data Transformation Pipelines, Feature Engineering Pipelines, Apache Airflow, Workflow Orchestration, Batch and Streaming Processing
- **Cloud Platforms & Infrastructure:** Amazon Web Services (S3, EMR, MSK, EC2, RDS, CloudWatch, Kinesis), Google Cloud Platform (BigQuery, Dataproc, Cloud Storage)
- **Databases & Data Warehousing:** PostgreSQL, BigQuery, MySQL, Amazon Redshift
- **Data Storage & Formats:** Parquet, Avro, JSON, Data Lake Architecture, Partitioned Tables, Columnar Storage
- **Data Analytics & Visualization:** Tableau, BigQuery Analytics, Marketing Attribution Analytics, Cohort Analysis
- **Monitoring & Observability:** Prometheus, Grafana, AWS CloudWatch, Logging and Pipeline Monitoring
- **Other Tools:** Redis, Elasticsearch, Linux, Git